

1. Matrix-free input-output (resolvent) analysis

$$\mathbf{u}_t + (\mathbf{U} \cdot \nabla \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{U}) = -\nabla p + \nabla^2 \mathbf{u} / Re. \quad (1.1a)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (1.1b)$$

The adjoint formulation is following (Parker *et al.* 2021; Biau 2016)

If we formally denote by $\mathcal{L}u = 0$ the linearised equations in (1.1), then the optimisation problem is to maximise the energy $E(t)$ subject to the constraint $\mathcal{L}u = 0$. The Lagrangian functional is based on an energy-like norm and reads

$$\mathcal{L}(u, u^\dagger) = E(t) + \int_V u^\dagger \cdot (\mathcal{L}u) dV,$$

with $u^\dagger$ the Lagrange multiplier or adjoint variable. The optimal state is reached for the stationary point for the Lagrange function, which gives us the adjoint problem:

$$\nabla \cdot \mathbf{u}^\dagger = 0, \quad (1.2a)$$

$$-\mathbf{u}_t^\dagger - (\mathbf{U} \cdot \nabla \mathbf{u}^\dagger - \mathbf{u}^\dagger \cdot (\nabla \mathbf{U})^T) = -\nabla p^\dagger + \nabla^2 \mathbf{u}^\dagger / Re \quad (1.2b)$$

The adjoint equations (1.2) are associated with the following boundary conditions: no-slip conditions at the wall,

$$\mathbf{u}^\dagger = 0,$$

For this term $\mathbf{u}^* \cdot (\nabla \mathbf{U})^T$, Kerswell *et al.* (2014); Kerswell (2018) write it $u_j \partial_i U_j$. This just switch the index i and j in the typical notation of $u_j \partial_j U_i$ meaning $\mathbf{u} \cdot \nabla \mathbf{U}$. Switching the index i and j is what transpose is doing.

Below will just write down the adjoint equations componentwise based on (1.2) and adding the body force penalty term, including 1D base flow, $U(y, z)$ base flow for snapwise wavy wall and $[U(x, y), V(x, y)]$ base flow for streamwise wavy wall.

For the penalty term, I think it will be the same in the adjoint and forward equations as that does not involve any gradient operators so it should be trivial.

1.1. 1D base flow

Forward equation:

$$\frac{\partial u}{\partial t} + U \partial_x u + v \frac{dU}{dy} = -\partial_x p + \nabla^2 u / Re - \frac{\Gamma}{\eta} u \quad (1.3)$$

$$\frac{\partial v}{\partial t} + U \partial_x v = -\partial_y p + \nabla^2 v / Re - \frac{\Gamma}{\eta} v \quad (1.4)$$

$$\frac{\partial w}{\partial t} + U \partial_x w = -\partial_z p + \nabla^2 w / Re - \frac{\Gamma}{\eta} w \quad (1.5)$$

$$\partial_x u + \partial_y v + \partial_z w = 0 \quad (1.6)$$

Adjoint equation:

$$-\frac{\partial u^\dagger}{\partial t} - U \partial_x u^\dagger = -\partial_x p^\dagger + \nabla^2 u^\dagger / Re - \frac{\Gamma}{\eta} u^\dagger \quad (1.7)$$

$$-\frac{\partial v^\dagger}{\partial t} - U \partial_x v^\dagger + u^\dagger \frac{dU}{dy} = -\partial_y p^\dagger + \nabla^2 v^\dagger / Re - \frac{\Gamma}{\eta} v^\dagger \quad (1.8)$$

$$-\frac{\partial w^\dagger}{\partial t} - U \partial_x w^\dagger = -\partial_z p^\dagger + \nabla^2 w^\dagger / Re - \frac{\Gamma}{\eta} w^\dagger \quad (1.9)$$

$$\partial_x u^\dagger + \partial_y v^\dagger + \partial_z w^\dagger = 0 \quad (1.10)$$

1.2. *y-z 2D base flow (spanwise wavy wall)*

$U(y, z)$.

Forward equation

$$\begin{aligned} \frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} + v \frac{\partial U}{\partial y} + w \frac{\partial U}{\partial z} &= -\frac{\partial p}{\partial x} + \nabla^2 u / Re - \frac{\Gamma}{\eta} u \\ \frac{\partial v}{\partial t} + U \frac{\partial v}{\partial x} &= -\frac{\partial p}{\partial y} + \nabla^2 v / Re - \frac{\Gamma}{\eta} v \\ \frac{\partial w}{\partial t} + U \frac{\partial w}{\partial x} &= -\frac{\partial p}{\partial z} + \nabla^2 w / Re - \frac{\Gamma}{\eta} w \\ \partial_x u + \partial_y v + \partial_z w &= 0 \end{aligned} \quad (1.11)$$

Adjoint equation:

$$\begin{aligned} -\frac{\partial u^\dagger}{\partial t} - U \frac{\partial u^\dagger}{\partial x} &= -\frac{\partial p^\dagger}{\partial x} + \nabla^2 u^\dagger / Re - \frac{\Gamma}{\eta} u^\dagger \\ -\frac{\partial v^\dagger}{\partial t} - U \frac{\partial v^\dagger}{\partial x} + u^\dagger \frac{\partial U}{\partial y} &= -\frac{\partial p^\dagger}{\partial y} + \nabla^2 v^\dagger / Re - \frac{\Gamma}{\eta} v^\dagger \\ -\frac{\partial w^\dagger}{\partial t} - U \frac{\partial w^\dagger}{\partial x} + u^\dagger \frac{\partial U}{\partial z} &= -\frac{\partial p^\dagger}{\partial z} + \nabla^2 w^\dagger / Re - \frac{\Gamma}{\eta} w^\dagger \\ \partial_x u^\dagger + \partial_y v^\dagger + \partial_z w^\dagger &= 0 \end{aligned} \quad (1.12)$$

1.3. *x-y 2D-2C base flow (streamwise wavy wall)*

$U(x, y)$ and $V(x, y)$.

NS perturbed momentum equation for channel flow with forcing term in streamwise direction with U and V mean flow

$$\begin{aligned} \frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} + V \frac{\partial u}{\partial y} + u \frac{\partial U}{\partial x} + v \frac{\partial U}{\partial y} &= -\frac{\partial p}{\partial x} + \nabla^2 u / Re - \frac{\Gamma}{\eta} u \\ \frac{\partial v}{\partial t} + U \frac{\partial v}{\partial x} + V \frac{\partial v}{\partial y} + u \frac{\partial V}{\partial x} + v \frac{\partial V}{\partial y} &= -\frac{\partial p}{\partial y} + \nabla^2 v / Re - \frac{\Gamma}{\eta} v \\ \frac{\partial w}{\partial t} + U \frac{\partial w}{\partial x} + V \frac{\partial w}{\partial y} &= -\frac{\partial p}{\partial z} + \nabla^2 w / Re - \frac{\Gamma}{\eta} w \\ \partial_x u + \partial_y v + \partial_z w &= 0 \end{aligned} \quad (1.13)$$

Adjoint equation

$$\begin{aligned}
-\frac{\partial u^\dagger}{\partial t} - U \frac{\partial u^\dagger}{\partial x} - V \frac{\partial u^\dagger}{\partial y} + u^\dagger \frac{\partial U}{\partial x} + v^\dagger \frac{\partial V}{\partial x} &= -\frac{\partial p^\dagger}{\partial x} + \nabla^2 u^\dagger / Re - \frac{\Gamma}{\eta} u^\dagger \\
-\frac{\partial v^\dagger}{\partial t} - U \frac{\partial v^\dagger}{\partial x} - V \frac{\partial v^\dagger}{\partial y} + u^\dagger \frac{\partial U}{\partial y} + v^\dagger \frac{\partial V}{\partial y} &= -\frac{\partial p^\dagger}{\partial y} + \nabla^2 v^\dagger / Re - \frac{\Gamma}{\eta} v^\dagger \\
-\frac{\partial w^\dagger}{\partial t} - U \frac{\partial w^\dagger}{\partial x} - V \frac{\partial w^\dagger}{\partial y} &= -\frac{\partial p^\dagger}{\partial z} + \nabla^2 w^\dagger / Re - \frac{\Gamma}{\eta} w^\dagger \\
\partial_x u^\dagger + \partial_y v^\dagger + \partial_z w^\dagger &= 0
\end{aligned} \tag{1.14}$$

REFERENCES

- BIAU, D. 2016 Transient growth of perturbations in stokes oscillatory flows. *Journal of Fluid Mechanics* **794**, R4.
- KERSWELL, R., PRINGLE, C. C. & WILLIS, A. 2014 An optimization approach for analysing nonlinear stability with transition to turbulence in fluids as an exemplar. *Reports on Progress in Physics* **77** (8), 085901.
- KERSWELL, R. R. 2018 Nonlinear nonmodal stability theory. *Annual Review of Fluid Mechanics* **50**, 319–345.
- PARKER, J., HOWLAND, C., CAULFIELD, C. & KERSWELL, R. 2021 Optimal perturbation growth on a breaking internal gravity wave. *Journal of Fluid Mechanics* **925**, A16.